

LETS AI - SARVAM

Yogesh Haribhau Kulkarni

Sarvam AI

India's AI Stack for Indian Languages

Sarvam AI: India's Language AI Platform

- ▶ **Mission:** Build AI that works for India — in Indian languages, with Indian accents, for Indian use cases.
- ▶ **Founded** by IIT Madras researchers; backed by Lightspeed, Peak XV (Sequoia India), and others.
- ▶ **22+ Indian languages** supported, covering all scheduled languages of the Indian Constitution.
- ▶ **Full AI stack:** Speech, Translation, LLM, OCR — a one-stop API platform for Indic language AI.
- ▶ Open-weights models (`sarvam-translate:v1`, Sarvam-M) available on HuggingFace.
- ▶ **Free credits:** Every new signup receives **1,000** worth of free API credits to get started immediately.

The Problem Sarvam Solves

- ▶ **India has 22 official languages** and 1.4 billion people — but most AI tools work only in English.
- ▶ Global models (GPT, Gemini) underperform on Indian languages, accents, and code-mixed text (“Hinglish”, “Tanglish”).
- ▶ Existing ASR systems fail on regional accents, noisy environments, and telephony-quality audio.
- ▶ **Sarvam trains on culturally grounded Indian data** for each language: not just translated English corpora.
- ▶ Practical impact: customer service bots, govt. scheme dissemination, edtech, fintech, healthcare — all in the user’s mother tongue.

Capability	Generic LLM API	Sarvam AI
Hindi/Gujarati/Marathi ASR	Mediocre	Best-in-class
22 Indian languages TTS	No	Yes
Code-mixed transcription	No	Yes (codemix mode)
Indian document OCR	No	Yes (23 languages)
-denominated pricing	No	Yes

The Sarvam AI Model Family

Model	Task	Languages	Status
Saaras v3	Speech → Text	23	✓ Recommended
Bulbul v3	Text → Speech	11	✓ Active
Mayura	Translation	11	✓ Active
Sarvam-Translate	Translation	23	✓ Active
Sarvam-30B	Chat LLM	11	✓ Active
Sarvam-105B	Chat LLM (flagship)	11	✓ Active
Sarvam Vision	Document OCR	23	✓ Active
Sarvam-M (legacy)	Chat LLM (24B)	11	Migrate away
Saarika v2.5 (legacy)	Speech → Text	11	Migrate away

All models trained on Indian language data; evaluated on Indian language benchmarks.

Language Coverage

Saaras v3 (STT) & Sarvam-Translate: All 23 languages

Language	Code	Language	Code
Hindi	hi-IN	Urdu	ur-IN
Bengali	bn-IN	Nepali	ne-IN
Kannada	kn-IN	Konkani	kok-IN
Malayalam	ml-IN	Kashmiri	ks-IN
Marathi	mr-IN	Sindhi	sd-IN
Odia	od-IN	Sanskrit	sa-IN
Punjabi	pa-IN	Santali	sat-IN
Tamil	ta-IN	Manipuri	mni-IN
Telugu	te-IN	Bodo	brx-IN
Gujarati	gu-IN	Maithili	mai-IN
English (India)	en-IN	Dogri	doi-IN
Assamese	as-IN		

Bulbul v3 (TTS) and Mayura (Translation) cover 11 languages (the 10 most-spoken Indian languages + English).

Key API Capabilities

- ▶ **Speech to Text (Saaras v3):** State-of-the-art ASR, 23 languages, 5 output modes: transcribe, translate, verbatim, translit, codemix. REST, Batch, WebSocket streaming.
- ▶ **Text to Speech (Bulbul v3):** 30+ speaker voices, 11 languages, customizable pitch/pace, 8+ audio formats. REST and WebSocket streaming.
- ▶ **Translation (Mayura / Sarvam-Translate):** Context-preserving translation with 3 register styles (formal, colloquial, general). Auto language detection.
- ▶ **Transliteration:** Script conversion preserving pronunciation. e.g., “namaste” ↔ “नमस्ते”. Supports spoken-form conversion.
- ▶ **Language Identification:** Detect language and script from raw text. 23 Indian languages + English.
- ▶ **Chat Completion (Sarvam-30B / 105B):** Multilingual LLM, 11 languages, OpenAI-compatible API.
- ▶ **Document Intelligence (Sarvam Vision):** OCR and structured extraction from PDFs in 23 languages. **Currently free.**

Pricing and Free Tier

Free on signup: 1,000 worth of credits. Credits never expire.

API	Price	Billing Unit
Chat Completion (Sarvam-M)	0	per token (free!)
Document Intelligence	0	per page (free!)
Speech to Text	30	per hour of audio
STT + Diarization	45	per hour
STT + Translate	30	per hour
Text to Speech (Bulbul v2)	15	per 10K chars
Text to Speech (Bulbul v3)	30	per 10K chars
Translation (Mayura / Sarvam-Translate)	20	per 10K chars
Transliteration	20	per 10K chars
Language Identification	3.5	per 10K chars

Plans: Starter (Pay-as-go, 60 req/min) | Pro (10,000/mo) | Business (50,000/mo) | Enterprise (custom).

Getting Started: Setup

```
1 # Step 1: Get your API key
2 # Sign up at dashboard.sarvam.ai
3 # Create an API key under "API Keys" section
4 # You receive Rs. 1000 free credits on signup
5
6 export SARVAM_API_KEY="your_api_key_here"
7
8 #Step 2: Install the Python SDK
9 pip install -U sarvamai
10
11 # Step 3: Make your first call (Translation)
12 from sarvamai import SarvamAI
13
14 client = SarvamAI(api_subscription_key="YOUR_API_KEY")
15
16 response = client.text.translate(
17     input="Hello, how are you?",
18     source_language_code="auto",    # auto-detect
19     target_language_code="hi-IN",  # translate to Hindi
20     speaker_gender="Female"
21 )
22 print(response)
23 # Output:      ,      ?
```

Speech to Text: Saaras v3

Five output modes — one model, many use cases:

- ▶ **transcribe**: Return text in the original spoken language (ASR).
- ▶ **translate**: Transcribe and translate to English in one step.
- ▶ **verbatim**: Word-for-word transcript with no post-processing.
- ▶ **translit**: Romanize the transcript (e.g., Hindi → Roman script).
- ▶ **codemix**: Handle code-mixed speech (“Hinglish”, “Tanglish”).

Supported audio formats: wav, mp3, aac, aiff, ogg/opus, flac, mp4/m4a, amr.

Max duration per REST request: **30 seconds**. Use Batch API for longer files.

```
1 from sarvamai import SarvamAI
  client = SarvamAI(api_subscription_key="YOUR_KEY")
3 # Transcribe a Hindi audio file
  response = client.speech_to_text.transcribe(
5     file=open("hindi_audio.wav", "rb"),
      model="saaras:v3",
7     mode="transcribe",    # or translate/verbatim/translit/codemix
      language_code="hi-IN" # or omit for auto-detection
9 print(response.transcript)
```

Text to Speech: Bulbul v3

30+ natural-sounding voices across 11 languages.

Sample voices:

- ▶ **Male (23):** shubh (default), aditya, rahul, anand, varun, kabir, mani, gokul, vijay ...
- ▶ **Female (16):** ritu, priya, neha, pooja, simran, kavya, ishita, amelia, kavitha ...

Output formats: mp3, wav, linear16, mulaw, alaw, opus, flac, aac.

```
1 from sarvamai import SarvamAI
  client = SarvamAI(api_subscription_key="YOUR_KEY")
3 response = client.text_to_speech.convert(
    target_language_code="hi-IN",
    text="      !                ?",
    model="bulbul:v3",
    speaker="anand", # Male voice
    # speaker="priya", # Female voice
    speech_sample_rate=24000, # 8000/16000/24000 Hz
    enable_preprocessing=True)
11 # Save the audio
  with open("output.wav", "wb") as f:
13     f.write(bytes(response.audios[0]))
```

Translation and Transliteration

Translation with register control:

```
1 response = client.text.translate(  
    input="Please submit your application form.",  
3    source_language_code="en-IN",  
    target_language_code="gu-IN",  
5    mode="formal",    # formal | colloquial | modern-colloquial  
    # formal: official docs/legal  
7    # colloquial: chatbots/social media  
    # modern-colloquial: business emails  
9    numerals_format="native"    # native (Gujarati numerals)  
                                # or international (0-9)  
11 )  
# Transliteration (script conversion, not translation):  
13  
# Hindi Devanagari -> Roman script  
15 response = client.text.transliterate(  
    input="      ",  
17    source_language_code="hi-IN",  
    target_language_code="en-IN",  
19    spoken_form=True,    # convert to spoken form  
    numerals_format="international"  
21 )  
print(response.transliterated_text)  
23 # Output: "namaste, mera naam vinayak hai."
```

Chat Completion and Language Identification

Available chat models:

- ▶ **sarvam-m**: 24B params, free — ideal for prototyping
- ▶ **sarvam-30b**: 30B params, balanced cost/quality
- ▶ **sarvam-105b**: 105B params, highest quality (flagship)

Chat Completion (OpenAI-compatible API):

```
1 response = client.chat.completions(  
    messages=[  
3         {"role": "system",  
           "content": "You are a helpful assistant. Reply in Hindi."},  
5         {"role": "user",  
           "content": "                ?"}],  
7     model="sarvam-m" # free! or sarvam-30b / sarvam-105b)  
print(response.choices[0].message.content)  
9  
# Language Identification:  
11 # Detect language and script from raw text  
response = client.text.identify_language(  
13     input="" )  
print(response.language_code) # "hi-IN"  
15 print(response.script_code) # "Deva" (Devanagari)
```

Document Intelligence: Free OCR

Use cases:

- ▶ Government forms, Aadhaar, PAN cards, bank statements in regional scripts
- ▶ Medical prescriptions written in Hindi/Gujarati/Marathi
- ▶ Legal documents, academic certificates
- ▶ Max 10 pages per request; rate limit: 10 req/min (all plans)

Extract structured text from PDFs in 23 languages. Currently free.

```
1 import base64, httpx
3 # Read and encode the PDF
4 with open("hindi_document.pdf", "rb") as f:
5     pdf_b64 = base64.b64encode(f.read()).decode()
7 response = httpx.post(
8     "https://api.sarvam.ai/v1/parse",
9     headers={"api-subscription-key": "YOUR_KEY"},
10    json={"file": pdf_b64, "file_type": "pdf" })
11 data = response.json()
12 print(data["text"]) # Extracted plain text
13 print(data["pages"]) # Page-by-page structured output
```

Real-World Use Cases

- ▶ **Multilingual Voice Bots:** Customer service bots that auto-detect Hindi/Gujarati/Marathi and respond in the same language. Used in banking, insurance, e-commerce.
- ▶ **Call Analytics:** Transcribe recorded calls with diarization (speaker labels), identify agent vs. customer, analyse sentiment. All in Indian languages.
- ▶ **Government Scheme Dissemination:** Text/voice agents that explain PMJAY, PM Kisan, etc. to rural users in their local language.
- ▶ **EdTech:** Multilingual tutors that can explain concepts in Hindi/Marathi/Kannada, accept student voice input, and respond in speech.
- ▶ **Fintech / Loan Advisory:** Vernacular loan advisors that guide borrowers through EMI calculations and documentation in regional languages.
- ▶ **Document Digitisation:** Extract and translate text from regional-language land records, certificates, and government documents.
- ▶ **Accessibility:** Screen readers and voice interfaces for users who are not literate in English.

Voice Agent Architecture

Building a full multilingual voice agent requires three layers:

Layer	Sarvam Model	Role
Speech Input	Saaras v3	Converts spoken Indian language to text
Understanding	Sarvam-30B / 105B	Processes query, generates response
Speech Output	Bulbul v3	Converts response text to natural speech

Integration frameworks supported:

- ▶ **LiveKit:** Real-time voice agents. Sarvam STT and TTS plugins available via `livekit-agents[sarvam]`.
- ▶ **Pipecat:** Pipeline-based voice agents. `SarvamSTTService` and `SarvamTTSService` in Pipecat SDK.
- ▶ **Direct WebSocket:** Real-time streaming STT and TTS via Sarvam's own WebSocket API (ultra-low latency).
- ▶ **REST API:** Simple synchronous calls for non-real-time pipelines.

Pro Tip: Use `language="unknown"` in Saaras STT to auto-detect the language. Essential for diverse Indian customer bases where users switch languages.

Batch Processing: Call Analytics

Pipeline: Batch STT → Parse speaker turns → LLM analysis → Sentiment/Resolution summary.

Batch STT API: Process up to 20 audio files (up to 1 hour each) asynchronously.

```
1 from sarvamai import SarvamAI
  client = SarvamAI(api_subscription_key="YOUR_KEY")
3 # Create a batch job with diarization
  job = client.speech_to_text_translate_job.create_job(
5     model="saaras:v3",
     mode="translate",      # translate to English
7     with_diarization=True, # speaker labels
  )
9 job.upload_files(file_paths=["call1.mp3", "call2.mp3"])
  job.start()
11 job.wait_until_complete() # blocking; checks status
   # Download results
13 job.download_outputs(output_dir="./transcriptions/")
   # Diarization output, labels each line by speaker:
15 SPEAKER_00: Hello, I'm calling about my EMI payment.
   SPEAKER_01: Sure, can I have your account number?
17 SPEAKER_00: It's 9876543210.
```

Summary

Why Use Sarvam AI?

- ▶ **India-first:** Models trained on Indian language data, not English translations. Handles real-world Indian accents, dialects, and code-mixing.
- ▶ **Full stack:** One API platform for Speech, Translation, LLM, and OCR — no need to stitch together multiple foreign vendors.
- ▶ **Generous free tier:** 1,000 free credits on signup. **Chat (Sarvam-M) and Document OCR are completely free.** No credit card required to start.
- ▶ **-denominated pricing:** Predictable costs in Indian Rupees; no USD conversion surprises. Competitive rates for Indian market scale.
- ▶ **Open weights:** sarvam-translate:v1 and Sarvam-M (based on Mistral-Small) available on HuggingFace for self-hosting.
- ▶ **Production-ready:** Three-tier rate limit system (Provisioned/Burst/High-Throughput), WebSocket streaming, Batch API, diarization.
- ▶ **Ecosystem:** LiveKit and Pipecat integrations; growing cookbook of real-world examples (call analytics, voice agents, document processing).
- ▶ **Bottom line:** If you are building for India, Sarvam AI gives you better accuracy, lower cost, and native language support that no global provider matches.

Quick Reference: API Cheat Sheet

Task	SDK Method	Model
Transcribe audio	<code>speech_to_text.transcribe()</code>	saaras:v3
Transcribe + translate	<code>speech_to_text.transcribe(mode="translate")</code>	saaras:v3
Code-mixed speech	<code>speech_to_text.transcribe(mode="codemix")</code>	saaras:v3
Batch transcription	<code>speech_to_text_translate_job</code>	saaras:v3
Text to speech	<code>text_to_speech.convert()</code>	bulbul:v3
Translate text	<code>text.translate()</code>	mayura / sarvam-translate
Transliterate	<code>text.transliterate()</code>	—
Detect language	<code>text.identify_language()</code>	—
Chat completion	<code>chat.completions()</code>	sarvam-m (free)
Document OCR	<code>/v1/parse (REST)</code>	sarvam-vision (free)

Key endpoints: <https://api.sarvam.ai/v1/>

Auth header: `api-subscription-key: YOUR_KEY`

Docs: docs.sarvam.ai · **Dashboard:** dashboard.sarvam.ai



- ▶ **Dashboard (API key + credits):** `dashboard.sarvam.ai`
- ▶ **Docs:** `docs.sarvam.ai`
- ▶ **API Reference:** `docs.sarvam.ai/api-reference-docs/introduction`
- ▶ **GitHub Cookbook:** `github.com/sarvamai/sarvam-ai-cookbook`
- ▶ **Sarvam GitHub Org:** `github.com/sarvamai`
- ▶ **API Status:** `status.sarvam.ai`
- ▶ **Discord Community:** `discord.gg/5rAsykttcs`
- ▶ **LiveKit Integration:**
`docs.livekit.io/agents/models/stt/plugins/sarvam`
- ▶ **Pipecat Integration:** `docs.pipecat.ai/server/services/stt/sarvam`
- ▶ **Python SDK:** `pip install sarvamai`
- ▶ **JS SDK:** `npm install sarvamai`
- ▶ **Support:** `developer@sarvam.ai`

Build for a billion Indians in their own languages.